



HOME / PROJECTS / TI PICCOLO PMSM DYNAMOMETER

MICHIGAN TECHNOLOGICAL UNIVERSITY RESEARCH AND GRADUATE LABORATORY PLATFORM

TI Piccolo PMSM *dynamometer.*

Developed in Herder Elektronische Systemen's lab using company-owned equipment, within Michigan Technological University's electrified-propulsion lab context, and prepared for integration into ME 5990 Electric Propulsion Systems graduate instruction and research in motor control, real-time systems, CAN communications, and drivetrain validation.

F28069M FIELD-ORIENTED CONTROL QEP CAN REAL-TIME TESTING

01 / CONTEXT

Michigan Tech *graduate lab context.*

Michigan Technological University's ME 5990 Electric Propulsion Systems provides the official course context for this platform. The dynamometer is prepared for integration into the graduate laboratory curriculum as a hands-on lab asset for advanced instruction and research in motor control, real-time systems, CAN communications, and drivetrain validation.

This work was completed in Herder Elektronische Systemen's lab using company-owned equipment. It is presented as public academic and research work associated with Michigan Technological University, not as a privately owned Herder Elektronische Systemen product or completed client engagement.

02 / ARCHITECTURE

Motors, controls *and evidence.*

Technical scope

The public material documents TI C2000 F28069M control, field-oriented control architecture, QEP encoder integration, CAN communications, Speedgoat/MATLAB real-time testing where documented, efficiency mapping and validation data processing.

SYSTEM

Dual-motor PMSM dynamometer integration.

METHODS

Torque-speed, power and efficiency validation workflows.

TESTBED ARCHITECTURE PLACEHOLDER

DUAL-MOTOR PMSM → F28069M → CAN → REAL-TIME TEST

Authorized dynamometer architecture diagram.

Source: public repository architecture documentation

SIGNAL MAP PLACEHOLDER

CAN AND QEP CHANNELS

CAN signal map or controller/testbed photograph.

MEASURED DATA PLACEHOLDER

SPEED / TORQUE / POWER

Measured plot with operating point and instrument context.

EFFICIENCY MAP PLACEHOLDER

VALIDATION RESULT

Efficiency map when source and publication authorization are confirmed.

PHASE SENSING, REGENERATION AND LONG-TERM TESTABILITY

Phase sensing, regeneration and *long-term testability.*

The phase-sense network was documented to preserve the dynamometer's operating knowledge and support future maintenance, troubleshooting and engineering use.

Each BOOSTXL-DRV8305EVM motor phase is sensed relative to ground through a 62.0 k Ω and 4.99 k Ω divider. Its ratio is approximately 0.07449 V/V, giving an expected VSEN_A, VSEN_B or VSEN_C range of 0 to 1.79 V from a 24 V DC bus.

A 0.1 μ F capacitor at each VSEN node forms a low-pass filter with the divider's approximate 4.62 k Ω Thevenin resistance. The resulting time constant is about 462 μ s and the cutoff frequency is approximately 344 Hz. The VSEN signals therefore represent filtered phase-to-ground switching behavior rather than sharp PWM waveforms.

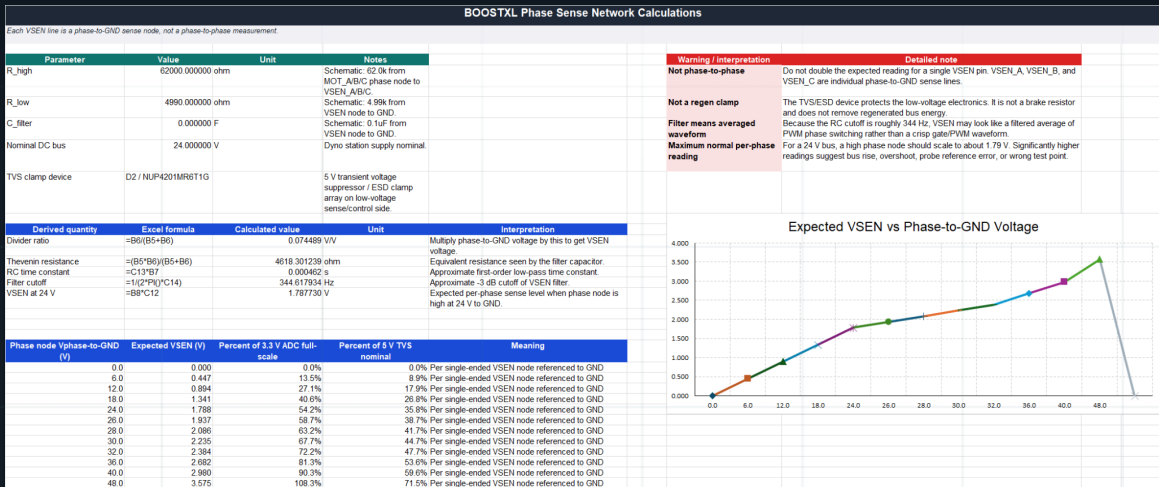
D2, an onsemi NUP4201MR6T1G TVS/ESD array, protects the low-voltage sensing electronics from electrostatic discharge and brief overvoltage transients. It does not absorb sustained regenerated energy or act as a brake resistor, DC-bus clamp or dump circuit.

During regenerative operation, the load motor returns energy through the inverter to the shared DC bus. If the supply cannot absorb that energy, the bus voltage can rise even though the VSEN inputs remain protected.

Regenerated energy must therefore be managed separately through a regenerative supply, controlled braking resistor, DC-bus clamp, temporary storage or operating limits appropriate to the final station architecture.

The phase-sense network supports measurement and diagnosis; the TVS array protects the sensing electronics; neither manages power returned to the DC bus.

Recording these boundaries helps future users distinguish sensing faults, ADC-range problems and transient events from genuine regeneration-driven bus overvoltage. It also preserves the assumptions and expected measurements needed to keep the dynamometer understandable, repairable and repeatable.



BOOSTXL phase-sense and protection calculation showing the 62.0 kΩ / 4.99 kΩ divider, 0.1 μF filtering, expected 0-1.79 V per-phase VSEN range at a 24 V bus, and the separate role of the NUP4201MR6T1G TVS/ESD array. The TVS protects low-voltage sensing electronics from transients; regenerated motor energy must be managed separately at the DC bus.

VSEN_A, VSEN_B and VSEN_C are individual phase-to-ground measurements. They are not direct phase-to-phase voltage signals.

03 / EVIDENCE

Measured work *with boundaries.*

Repository documentation, hardware integration notes, MATLAB/Python processing and measured plots support the scope described here. Complete worked examples, raw Simulink models, sponsor material and credentials remain outside this site.

VIEW REPOSITORY ↗

RELATED SERVICE: TEST AUTOMATION →

[O](#) [START A TECHNICAL DISCUSSION](#)

Bring the *technical question.*

Describe the system, its current stage and the evidence available. Secure file exchange can follow the initial discussion.

[START A TECHNICAL DISCUSSION →](#)

Or write directly: [Email us](#)



HERDER
ELEKTRONISCHE SYSTEMEN

© 2026 Herder Elektronische Systemen

[Privacy](#) [Engagement notes](#) [All projects](#)